Efficient automatic traffic light alteration for sparse traffic flow

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Abstract

This paper presents proof of concept for a novel traffic light control system which reduces average waiting times along with vehicle emissions. Cameras monitor arriving vehicles and pedestrians on all roads leading to the junction, arrival times are predicted, and these are used to change the traffic lights in favour of an approaching vehicle if there is no contention (i.e. no other vehicle or pedestrian approaching from another direction).

Keywords: Computer Vision Application, Vehicle and pedestrian detection, Optical Flow Estimation

1 Introduction

While some research has been performed on optimising traffic flow at busy junctions, very little consideration has been given to junctions with sparse traffic flows and little consideration seems to have been given to pedestrians. The work presented in this paper considers junctions with relatively small amounts of traffic, and also considers pedestrians when deciding how to set the traffic lights. The system tracks all vehicles and pedestrians on all roads and pavements leading to the junction and predicts their arrival times. Based on these estimates, in low traffic volume situations, the traffic lights at the junction can be set to prioritise the first vehicle/person to arrive at the junction, in this way eliminating unnecessary slowing/stopping time. In higher volume traffic situations the traffic lights can operate in the normal fashion.

2 State of the Art

A number of technologies exist for monitoring traffic, the main ones being inductive loops, video imaging and microwave radars. Inductive loop sensors consist of a coil which is excited by an oscillator, this causes it to generate a magnetic field in its surrounding area, this magnetic field resonates at a constant frequency which is monitored by an electronic detector. However, due to the nature of how inductive loops work, they are incapable of detecting the presence of pedestrians and as they are usually placed about 2 meters from the stop line they cannot be used to predict arrival times.. Microwave radars can detect vehicles and pedestrians, and are robust against lighting and weather variations. However, they are unable to detect stopped objects, which can be problematic in traffic-centric applications [Bernas et al., 2018].

Video imaging has been used extensively in traffic detection; for example, used to detect vehicles, count traffic flow to detect when vehicles break red lights, etc. [Guo et al., 2019] proposed a reinforcement learning based approach to traffic signal control at urban intersections, using SUMO to simulate different traffic patterns and junctions to improve waiting times. [Osman et al., 2017] describe a system for using cameras to determine traffic density on roads leading to a junction. The cameras have precalculated viewing angles and while the system is intended to optimise the amount of waiting time, this is specifically aimed at busy junctions, but no consideration is given to pedestrians. [Abbas et al., 2024] describe a system with high accuracy at vehicle detection under varying conditions and alter the green signal duration based on traffic volume. Again this system is targeted at high volume roads and no consideration is given to pedestrians.

The demand for adaptive traffic control systems is rising due to the increasing number of vehicles worldwide.

As of 2022 the SCATS system held the largest market share. SCATS uses information from vehicle detectors located in each lane at the stop line to adjust signal timings in response to variations in traffic demand and system capacity. It does not perform network-wide optimization but instead acts as a heuristic feedback system, adjusting timings based on changes in traffic flow from previous cycles [Hodgkiss and Ta, 2020]. While SCATS can reduce delays in low-traffic conditions, its reliance on data from prior cycles rather than the current moment means its response is not truly real-time, which can limit effectiveness when conditions change suddenly.

Pedestrian considerations are rare in adaptive traffic control research. In the survey by [Qadri et al., 2020], the authors note that most studies focus on traffic signal timings and their effects on vehicle delay and emissions, with little attention to pedestrian needs. One example is the work by [Yu et al., 2017], which models pedestrian delays under high pedestrian demand and unsaturated vehicle traffic, factoring in crossing type and vehicle behavior. However, this work only optimizes static signal timings based on historical data and does not incorporate real-time pedestrian or vehicle data.

3 Video Processing

We use cameras looking at each road approaching the junction, and in each video feed identify road areas and pavement areas, identify vehicles and pedestrians approaching the junction and then predict when they will arrive so that the traffic light can be changed/set optimally.

3.1 Identifying road and pavement regions

Cameras mounted on the traffic lights at intersections can view a specific area of roadway and pavement. To detect vehicles, cyclists and pedestrians approaching the intersection YOLOv8 is used. However, stationary vehicles parked near the intersection can be mistakenly classified as approaching traffic, and vehicles traveling in the opposite direction may also trigger false positives when determining vehicles that are relevant to the current signal phase. To remedy this, Farnebäck dense optical flow algorithm is used to estimate the motion of each detected object between consecutive frames.

Figure 1: Regions identified for vehicles (on the left in red) and for pedestrians (on the right in blue).

The detected vehicles and pedestrians are accumulated over a set period of time to generate a heatmap of active road and pedestrian areas, following the generation of the heatmap, thresholding is applied to this heatmap to define Regions of Interest for the road and pavement. Thresholding is performed by discarding all road and pedestrian regions with values below the median for vehicles and pedestrians, respectively, on their heatmaps (See Figure 1). This helps remove one-off detections, such as vehicles backing out of driveways or pedestrians crossing the road at undesignated areas. Note that whether a bicycle is treated as a vehicle or a pedestrian depends on whether it appears on the road or on the pavement (See Figure 2).

Figure 2: Whether a bicycle is treated as a vehicle or a pedestrian depends on whether it appears on the road or the pavement.

3.3 Predicting arrival times at the junction

To estimate the time remaining before a vehicle reaches the intersection, a Long Short-Term Memory (LSTM) neural network is used. For the use-case of traffic, vehicular motion across frames can be considered as sequential where small variations in velocity or position may indicate acceleration or deceleration. The model is trained using a sequence of 5 consecutive frame-wise feature vectors collected for each tracked vehicle, consisting of the bounding box location, the velocity of the bottom right corner together with the width & height of the bounding box. Each input sequence corresponds to a unique vehicle, and the output is a single scalar: the number of frames remaining before the vehicle reaches the intersection. See Figure 3. The predicted arrival time for a pedestrian should be computed in a similar manner although probably over a longer number of frames, or by reducing the frame rate (as pedestrian exhibit slower and jerkier motion, relative to vehicles).

3.4 Evaluation

Our dataset for testing consisted of a 20-minute test video which was split into two parts. 80% was used for training and 20% for testing. The prediction of when the vehicles would arrive at the junction was compared with the actual arrival times. The mean average error was 20 frames (equivalent to 0.66s).

Figure 3: Changing predictions for the number of frames for the lead car to the traffic lights (90 frames to the traffic lights in the left image, and 36 frames to the traffic lights in the right image).

4 Traffic Simulation

Testing the system in a real-world traffic intersection poses significant safety risks since it involves altering real traffic lights. Hence to evaluate the effectiveness of the proposed methodology, a simulation based approach is adopted. Since the proposed system focuses on predicting the precise arrival time of detected vehicles and dynamically adjusting the traffic signals, it requires fine-grained analysis on an individual vehicle level. We make use of the Simulation of Urban Mobility (SUMO) model to simulate the behaviour and motion of individual vehicles in terms of position, speed and acceleration [Lopez et al., 2018].

4.1 Intersection Design

A custom intersection was designed in SUMO which consisted of a crossroads (where four roads meet together). Each road has two-way traffic, designated cycle lanes and a pedestrian crossing at the intersection. To simulate the prediction model (indicating when vehicles will arrive at the junction), SUMO's Traffic Control Interface (TraCI) is used. This provides the distance to the intersection and the speed of each vehicle which can used to predict arrival times. A similar approach is used to detect the presence of pedestrians.

Vehicles are introduced into the simulation dynamically using a stochastic process that introduces variability in both the route choices and the departure intervals. For each vehicle, a random route is selected from the 12 possible routes (as vehicles come along 1 road out of 4 and may continue through the junction in 1 of 3 ways) and a departure time is based on a Gaussian distribution with a set mean interval and standard deviation of 5 seconds. To examine how the system performs across varying traffic intensities, the mean interval is varied across experiments to range from 1 to 18 seconds, effectively simulating traffic flow rates from 3 vehicles/minute to 60 vehicles/minute. This allows for the analysis of how congestion and emissions react to varying traffic conditions.

4.2 Default System

In order to observe the effects of the adaptive traffic light system, the generated traffic was also tested by using a standard timer based schedule. This default schedule assigns predetermined green, yellow and red light durations, while yellow light durations were kept consistent across both adaptive and the default schedule, the green and red light timings for the static schedule were determined using experimental evaluation to ensure a reasonable timing was set according to low traffic scenarios. For the simulated intersection used in this study, the most optimal static timings were found to be 20 seconds of green for each pair of parallel lanes, followed by 20 seconds for all four pedestrian crossings

4.3 Evaluation

SUMO can observe each vehicle in the simulation at a granular level, which allows for the extraction of detailed performance metrics such as individual and aggregated waiting times (i.e. the total duration a vehicle remains stationary due to red lights or congestion). The total combined waiting time of all vehicles passing through the intersection is calculated for each simulation run. The set of vehicles, their routes and frequency remain consistent across experiments when making comparisons to ensure a fair evaluation. See Figure 4.

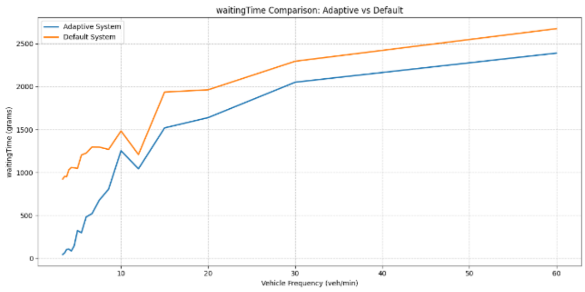

Figure 4: Comparison of the default traffic light control system vs. our adaptive system in terms of absolute average waiting time.

5 Conclusions

This paper has presented a proof of concept for traffic lights with sparse traffic flow, demonstrating separately that vehicular and pedestrian traffic can be located visually and that by predicting when that traffic will arrive at the junction, average waiting time along with emissions can be significantly reduced. Where the number of vehicles per minute was less than 7 the average waiting time dropped by more than 50% while at the same time emissions dropped by more than 20%.

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